

# Brandon Li

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## EDUCATION

**University of California, Los Angeles (UCLA)**

*Bachelor of Science in Computer Science*

Expected June 2028

*Los Angeles, CA*

## EXPERIENCE

**Software Development Engineer Intern**

*Amazon (AWS)*

Sep 2026 – Present

*Boston, MA*

**Software Engineering Intern**

*Google (YouTube)*

June 2026 – Sep 2026

*San Bruno, CA*

- Engineered 10-bit software AV1 HDR video playback for the YouTube Android app via low-level ARM/NEON SIMD vectorization in C++ to improve the video quality and playback performance for millions of users.
- Defined per-device video resolution caps (e.g. 1080p) for dav1d software decoding, filtering higher resolution formats out of the playback selection to prevent thermal throttling and battery drain on lower-end devices.
- Built a Java-based shader effect debug UI and OpenGL video pipeline to accelerate developer iteration speed.
- Integrated custom fragment shaders to allow video effects across thousands of DRM-protected shows and movies.
- Resolved memory leaks, migrated deprecated code, and improved format filtering across ExoPlayer and YouTube.

**Open Source Contributor**

*VideoLAN*

Jan 2026 – June 2026

*Remote*

- Built a shared dynamics core powering limiter and expander audio filters to cut duplicate code by 20%.
- Integrated ggmml and OpenCV dependencies into VLC's build system to enable on-device model inference.
- Ported 3 legacy OpenCV modules from C to C++ and upgraded the SAM2 filter to SAM3 to allow text prompts.
- Cut the final VLC binary size by 5% using linker-level dead-code stripping to reduce unnecessary dependencies.

**Software Engineering Intern**

*VideoLAN*

June 2025 – Sep 2025

*Remote*

- Integrated a SAM2 model module with OpenCV to enable real-time segmentation of 100+ concurrent objects.
- Implemented YuNet-based ML face detection to allow tracking for 100+ concurrent faces in surveillance systems.
- Built fast video-to-GIF conversion with five dithering algorithms for customizable quality and performance.

## PROJECTS

**yt-media-storage** | C++, Assembly/SIMD, FFmpeg, Coding Theory, Compression

Feb 2026 – Present

- Built an C++ tool (910 stars, 110+ forks) that encodes files into YouTube videos for storage.
- Achieved a 100x speed-up using inline Assembly, SIMD intrinsics, and OpenMP parallelization.
- Designed lightweight Qt 6 UI and libsodium encryption, enhancing user experience and security.
- Uses robust CRC and Wirehair error-correction to bypass lossy video compression algorithms.
- Reached #5 on Hacker News (Y Combinator) within 12 hours, generating thousands of views.
- Includes [technical explanation video](#) with over 3.4M impressions, educating over 280k viewers.

**mcav** | Java, Spring Boot, TypeScript, CI/CD

June 2020 – Present

- Built an 12-module, 150K-line Java media framework (130+ stars, 10 forks) whose unified player API streams real-time video from VLC, OpenCV, Chromium, VNC, and QEMU sources across four pluggable render backends.
- Designed a pure-GPU (GLSL fragment shader) video codec with O(1) per-pixel random access, improving BD-rate over 40% with VMAF-scored optimization rounds to reach 10 ms/frame at 1080p for integrated graphics cards.
- Engineered a low-latency pipeline API with nearly 50 different documented video effects that can be applied.
- Self-hosted a TeamCity CI/CD pipeline publishing releases to a Maven repository, gated with 2,500+ tests.

## TECHNICAL SKILLS

**Languages:** Java, C/C++, OCaml, JavaScript/TypeScript, Assembly

**Frameworks & Libraries:** Spring, Hibernate, FFmpeg, OpenCV, libVLC, React, Next.js, Tailwind

**Technologies:** Git, MongoDB, AWS, Google Cloud, Oracle Cloud, CMake, Meson, OpenGL, ExoPlayer, Android

**Awards:** Dean's Honor List